

# Junghun Hwang (황정헌)

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## Education

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### Konkuk University

Mar 2022 - Expected Feb 2027

B.S. in Smart Mobility Engineering | Seoul, Korea

**Completed coursework:** Linear Algebra, Autonomous System Platform, Artificial Intelligence Applications, Data Structures and Algorithms

**In progress:** AI Mobility Systems

## Research Interests

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Autonomous-driving perception and decision-making; deep learning and end-to-end driving, with an interest in learning-based policies and closed-loop evaluation.

## Research & Project Experience

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### 2026 9th Kookmin University Autonomous Driving Competition - AI-Edge Challenge

Apr 2026 - Aug 2026

Team Lead | Team of 5

- Built a ROS 2 imitation-learning workflow from data collection and training to TorchScript inference; implemented switching between learned steering and rule-based mission logic.
- Prepared and reviewed Roboflow datasets and trained a unified YOLO detector for traffic lights, vehicles, obstacles, and cones.
- Deployed Mission Manager / Final Driver integration and tuned the competition vehicle. The final driving strategy used rule-based control after steering oscillation persisted in the learned policy.
- Defined parking waypoints on a SLAM map and implemented reverse recovery logic for the parking event.

Team results: Preliminary 2nd Place | Finals 13th Place (course completed) | Parking Special Award

### 2026 Student AI · SW Mobility Competition

Mar 2026 - Ongoing

Team of 27 | Decision-Making Subteam (5)

- Developed ROS 2 AEB and mission logic with cone-boundary path generation, red-entry stopping, and stateful fail-safe behavior; implemented Stanley tracking with steering limits and motion-preview compensation.
- Validated the AEB pipeline in Gazebo closed-loop simulation and inspected path, steering, and stop states in Foxglove.
- In real-vehicle practice at the competition venue (Sep 19-20), completed waypoint runs on both preliminary and final courses at 10 m/s (36 km/h); verified AEB stop-command issuance at 50 km/h, while a brake hardware fault prevented a stop within the required zone.

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## Camera-Based V2I Cooperative End-to-End Autonomous Driving

2026 - Ongoing

Capstone Design | Requirements & System Design; Implementation Planned

- Defined a daytime, low-speed pedestrian-occlusion ODD for a 1/5-scale HENES platform and a matched comparison of camera-only and V2I-assisted E2E driving.
- Specified a road-plane homography pipeline and direct transmission of pedestrian position, confidence, and timestamps for lightweight fusion into waypoint and target-speed outputs.
- Formulated a detection-distance budget using latency, braking distance, and safety margin; planned stagewise latency and perception/deceleration lead measurements, with minimum-TTC and collision evaluation.

## ROS 2 UAV Waypoint Mission on PX4 SITL/Gazebo

May 2026 - Jun 2026

Autonomous System Platform Course | UAV Waypoint Development | Team of 4

- Developed a ROS 2 package that loads 3D waypoints from CSV and advances goals using TF-based pose tracking and configurable arrival tolerances.
- Connected the mission node to an existing PX4 offboard controller using PoseStamped commands, with coordinate-mode handling and mission lifecycle logic.
- Demonstrated sequential waypoint tracking and final autonomous landing in the PX4 SITL/Gazebo course simulation.

## V-Model Neuro-Symbolic Driving

Mar 2026 - Jun 2026

Dream Semester Project | Decision-Making | Team of 3

- Implemented ROS 2 decision-making, Pure Pursuit waypoint tracking, and longitudinal control with CARLA integration and later vehicle-interface adaptation.
- Investigated unstable simulated tracking by adjusting steering direction, lookahead, target smoothing, and command limits; the root cause remained unresolved.
- The autonomous-driving objective was not achieved, and the final code did not lead to an actual vehicle run. The work informed later interface and control-integration debugging.

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## Awards & Competitions

### 2026 9th Kookmin University Autonomous Driving Competition - AI-Edge Challenge

2nd Place, Preliminary Round | 13th Place, Finals | Special Award - Parking Event

### 2022 Drone Clash, General Division

3rd Place

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## Technical Skills

**Languages:** Python, C++

**Autonomous Systems:** ROS 2, CARLA, Gazebo, PX4 SITL

**AI / Computer Vision:** PyTorch, YOLO, OpenCV, Roboflow

**Tools / Environment:** Ubuntu, Git/GitHub, Foxglove, RViz2, rosbag2, QGroundControl